7 **(a)** **(i)** – 5.17 + 3.50 eV = –1.67 eV => Highest energy level is n = 3

number of spectral line = ${}^{3}C_{2} = 3$ [A1]

(ii) $hf = E_2 - E_1$

$(6.63\times10^{-34})f = \left(-3.07-(-5.17)\right)\left(1.60\times10^{-19}\right)$ [C1]

$f = 5.07\times10^{14}$ Hz [A1]

(iii) 1. from ground state to n = 2: <u>cannot occur</u> [B1]

2. from ground state to n = 3: <u>can occur</u> [B1]

The energy of the photon must be exactly <u>equal</u> to the <u>energy difference between two energy levels</u> before it can be absorbed. [B1]

(iv) There are <u>discrete energy levels</u> within the <u>nucleus</u>. [B1]

The energy <u>transitions are of large energies</u>, since the radiation is in the gamma range. [B1]

(b) **(i)** When high speed electrons emitted from the decay of Yttrium <u>**collides** with the **lead** nucleus/ are deflected by **lead** nucleus</u> [B1],

it undergoes <u>**large** deceleration/</u>slows down <u>abruptly</u>. This produces X-ray photons [B1].

The X-ray photons produced are of a range of energies as the electrons <u>decelerates to **different** extent</u>. [B1]

(ii) $E = \dfrac{hc}{\lambda}$

$\left(100\times10^{3}\right)\left(1.60\times10^{-19}\right) = \dfrac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{\lambda}$ [C1]

$\lambda = 1.243\times10^{-11}$ m

$p = \dfrac{h}{\lambda}$

$= \dfrac{\left(6.63\times10^{-34}\right)}{\left(1.243\times10^{-11}\right)}$ [C1]

$= 5.3\times10^{-23}$ kg m s^{-1} [A1]

(iii) $\text{Force} = \dfrac{\left(1.6\times10^{5}\right)\left(5.3\times10^{-23}-0\right)}{1}$ [C1]

$= 8.5\times10^{-18}$ N [A1]

(c) Photoelectric experiment. [B1]

Discussion of the evidence:

1) The <u>existence of a threshold frequency</u> below which no photoelectrons are emitted proves that <u>electromagnetic radiation</u> (EM) <u>consists of discrete quanta of energy given by *hf*</u>. [B1]

2) The instantaneous emission of photoelectrons when all the photon energy is transferred immediately to the electron in a single collision gives evidence to particulate nature of EM. [B1]

3) The maximum kinetic energy of the photoelectrons being dependent only on frequency of radiation *f*, which relates to the discrete energy of photon, and independent on the intensity of radiation also give evidence for the particulate nature of EM. [B1]